

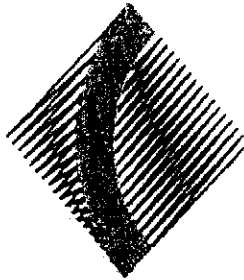
AT
KW
LB
PV
FH
AW

Name: _____

Class: 12MT2__ or 12MTX__

Teacher: _____

CHERRYBROOK TECHNOLOGY HIGH SCHOOL



2005

YEAR 12

AP4 EXAMINATION

MATHEMATICS

*Time allowed - 3 HOURS
(Plus 5 minutes reading time)*

DIRECTIONS TO CANDIDATES:

- Attempt all questions.
- All questions are of equal value.
- Each question is to be commenced on a new page clearly marked Question 1, Question 2, etc on the top of the page. **
- Each question is to be returned in a separate bundle.
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used.

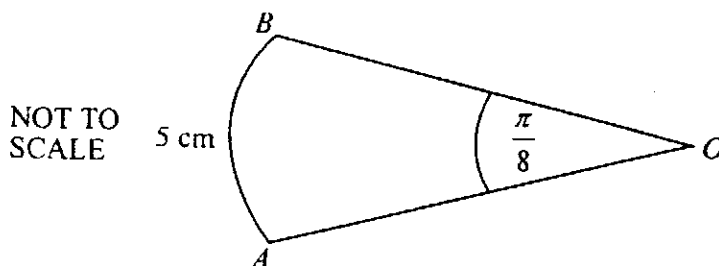
*****Each page must show your name and your class. *****

QUESTION ONE.**(12 MARKS.)****MARKS**

- (a) Evaluate $\sqrt{\frac{e^3 - 2.3}{3\pi}}$ correct to 3 significant figures. 2
- (b) Completely factorise $x^3 + 2x^2 - 8x$. 2
- (c) Differentiate with respect to x :
- (i) $3x^2(4x - 1)$ 2
- (ii) $\frac{\log_e x}{x^3}$ 3
- (d) Solve the pair of simultaneous equations: 2
- $$\begin{aligned}x - 2y &= 9 \\ 2x + y &= 8\end{aligned}$$
- (e) Find the primitive of $\frac{1}{x-1}$. 1

QUESTION TWO.**START A NEW PAGE.****(12 MARKS)**

- (a) In the diagram, AB is an arc of a circle with centre O. The length of the arc AB is 5cm and angle AOB is $\frac{\pi}{8}$ radians. Find the length of AO, to the nearest cm. 2



- (b) Solve $2^{2x} - 9 \cdot 2^x + 8 = 0$ 2
- (c) D(0,-2), E(4,0) and F(2,4) are three points on the number plane.
- (i) Draw a diagram to represent this information. 1
- (ii) Calculate the length of the interval DF. 1

QUESTION TWO CONTINUED.

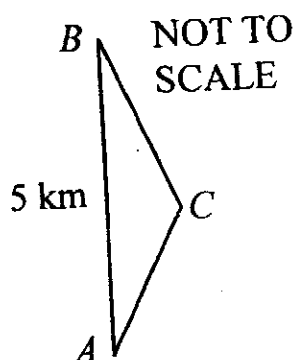
	MARKS
(iii) Calculate the gradient of DF.	1
(iv) Write the equation of the line DF in general form.	1
(v) Calculate the perpendicular distance from E to the line DF, giving your answer with a rational denominator.	2
(vi) Calculate the area of the ΔDEF .	2

QUESTION THREE.

START A NEW PAGE. (12 MARKS)

- (a) Differentiate $\frac{\tan 3x}{3x+2}$. 2
- (b) Find $\int \frac{3}{\sqrt{e^x}} dx$. 2
- (c) Evaluate $\int_2^4 \frac{3x}{x^2-1} dx$ leaving your answer exact and in its simplest form. 3

(d)



The buoy B is directly north of the buoy A. Because of the direction of the wind, to sail from A to B a boat must first sail from A to C on a bearing 025° and then turn and sail from C to B on a bearing of 335° .

If A and B are 5km apart,

- (i) find the size of $\angle ABC$. 1
- (ii) Hence, find the distance from A to C, 2
- (iii) Calculate the total distance sailed in going from A to B via C, to the nearest km. 2

QUESTION FOUR.**START A NEW PAGE.****(12 MARKS)****MARKS**

- (a) Solve $4 \cos x = 1$ for $0 \leq x \leq 2\pi$. Express your answer in radians, correct to 2 decimal places. 2
- (b) The quadratic equation $2x^2 + 9x - 4 = 0$ has roots of α and β . Hence evaluate $\alpha^2 + \beta^2$ 2
- (c) Shopping trolleys are 80cm long. When two trolleys are pushed together the resulting pair is 100cm long. When three are pushed together the resulting triple is 120 cm long.
- (i) How long will a set of 10 trolleys pushed together be? 2
- (ii) For safety reasons trolley collectors are not allowed to push a set of trolleys more than 4.5 metres long. What is the greatest number of trolleys that can be pushed? 2
- (d) For the parabola $4x = 8y - y^2$,
- (i) Find the coordinates of the vertex. 2
- (ii) Find the coordinates of the focus. 1
- (ii) Sketch the curve clearly labeling the vertex and focus. 1

QUESTION FIVE.**START A NEW PAGE.****(12 MARKS)**

- (a) Consider the function defined by $f(x) = 2x^3 - 3x^2 - 36x + 26$
- (i) Find the coordinates of the stationary points of the curve $y = f(x)$ and determine their nature. 3
- (ii) Find the coordinates of any point of inflection. 2
- (iii) Sketch the graph of $f(x) = 2x^3 - 3x^2 - 36x + 26$ by showing the above information. 2
- (iv) For what values of x is the curve concave down and decreasing? 2

QUESTION FIVE CONTINUED.**MARKS**

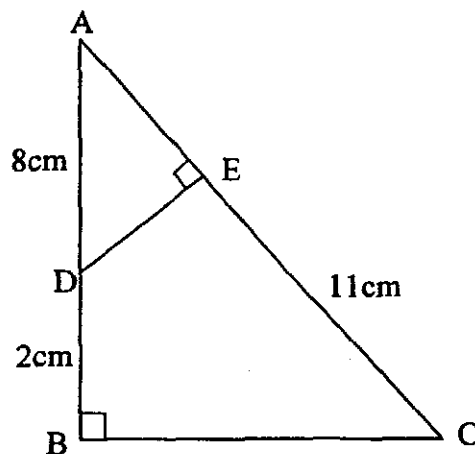
- (b) Two-digit numbers are formed from the digits 2, 3, 4, 6 with no repetition of digits allowed.
- (i) Write down the set S of all possible outcomes.
- (ii) A two-digit number is then selected at random from the set S. Find the probability that the number is prime.

2

1

QUESTION SIX.**START A NEW PAGE.****(12 MARKS)**

- (a) ABC is a right-angled triangle in which, $\angle ABC = 90^\circ$. Points D and E lie on AB and AC respectively such that DE is perpendicular to AC. AD = 8cm, EC = 11cm, and DB = 2cm.



Not to Scale

- (i) Prove that $\triangle ABC$ is similar to $\triangle AED$.
- (ii) Find the length of AE.

3

1

QUESTION SIX CONTINUED.

MARKS

(b) Consider the geometric series : $1 + (5 - \sqrt{p}) + (5 - \sqrt{p})^2 + (5 - \sqrt{p})^3 + \dots$

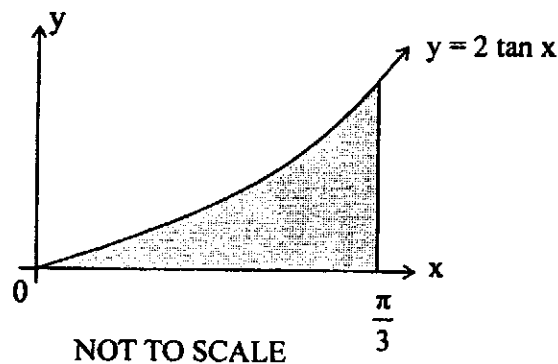
(i) Find the values of p for which the geometric series has a limiting sum?

2

(ii) Find the limiting sum of the series given that p is 20.
write your answer with a rational denominator.

2

(c) The shaded area between the curve $y = 2 \tan x$, the x -axis and the line $x = \frac{\pi}{3}$ is rotated about the x -axis.



Calculate the volume of the solid formed.

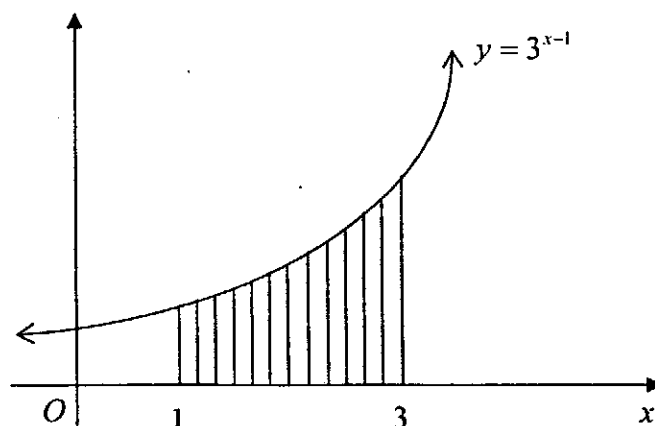
4

QUESTION SEVEN.

START A NEW PAGE.

(12 MARKS)

(a) The diagram below shows the shading of a region bounded by the graph $y = 3^{x-1}$ and the lines $x = 1$ and $x = 3$.



QUESTION SEVEN CONTINUED.

MARKS

- (i) Copy and complete the following table giving your answer correct to three decimal places:

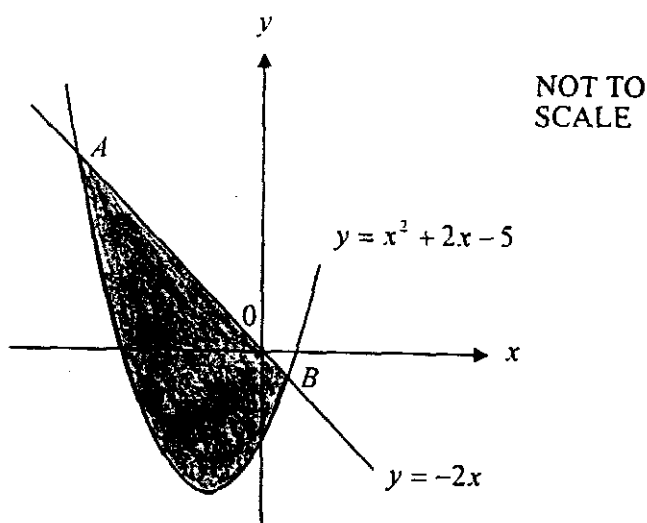
1

x	1	1.5	2	2.5	3
$y = 3^{x-1}$	1	1.732			

- (ii) Use Simpson's Rule with five function values to approximate the shaded area to three decimal places.

2

(b)



The diagram shows the graphs of $y = x^2 + 2x - 5$ and $y = -2x$. These two graphs intersect at the point A and B.

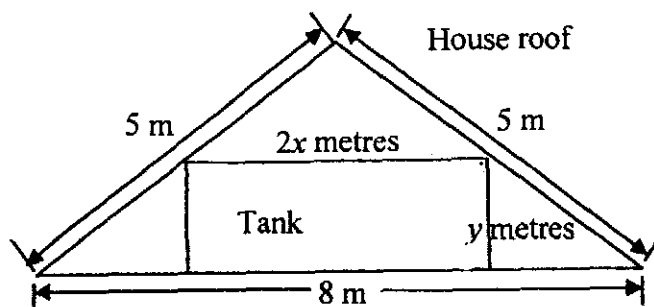
- (i) Find the x values of the points of intersection A and B 2
- (ii) Calculate the area of the shaded region. 3
- (c) Write down the equation of the tangent to the curve $y = e^{2x}$ at the point where $x = 1$. 2
- (d) Evaluate $\sum_{n=2}^4 -2^n$ 2

QUESTION EIGHT.**START A NEW PAGE.****(12 MARKS)****MARKS**

- (a) The probability of winning a particular game of chance is 20%.
- (i) What is the probability of winning all three games? 1
 - (ii) If she plays three games, what is the probability she will win at least one? 2
 - (iii) What is the least number of games she must play before the probability of her winning at least one game is 99% 3
- (b) When Penny began her first job at Hornsby, she decided to open an account and deposit \$50 per month into the account. The bank guaranteed an interest rate of 8%p.a. compounding every 6 months if she agreed not to make any withdrawals.
- (i) How much will be in this account at the end of 12 months? 1
 - (ii) How much will be in this account at the end of 30 years? 2
 - (iii) How much will be in this account at the end of 30 years if Penny doubles the amount of her monthly deposit to \$100 per month at the beginning of the 11th year, and doubles again to \$200 per month at the beginning of the 21st year? 3

QUESTION NINE.**START A NEW PAGE.****(12 MARKS)**

- (a) The diagram below shows the cross section of a cylindrical hot water tank, with diameter $2x$ metres and height y metres that fits exactly into the roof of a house. The cross section of the roof is an isosceles triangle with the base 8 metres and equal sides 5 metres in length.

**NOT TO
SCALE**

QUESTION NINE CONTINUED.**MARKS**

- (i) Explain why the roof is 3 metres high? 1
- (ii) By using similar triangles, show that $y = \frac{3}{4}(4 - x)$. 2
- (iii) Show that the volume of the tank, V metres³, is given by 1
- $$V = \frac{3\pi}{4}(4x^2 - x^3)$$
- (iv) Use calculus to find the radius of the tank that gives it a maximum volume. 3
- (v) Calculate the maximum volume. 1
- (b) (i) If the line $y = x + m$, cuts the circle $x^2 + y^2 = 4$, show that the x coordinates of the points of intersection can be found by solving $2x^2 + 2xm + m^2 - 4 = 0$ 2
- (ii) For what values of m will the line $y = x + m$, be a tangent to the circle. 2

QUESTION TEN.**START A NEW PAGE.****(12 MARKS)**

- (a) (i) Simplify $\log_e e^{2ax}$ 1
- (ii) Hence evaluate $\int_a^b \log_e e^{2ax} dx$ 2
- (b) Express $\frac{\cos^3 \theta}{\sin \theta} + \sin \theta \cos \theta$ as a single trigonometric ratio. 2
- Hence, solve $\frac{\cos^3 \theta}{\sin \theta} + \sin \theta \cos \theta = 1$ for $0 \leq \theta \leq 2\pi$.

QUESTION TEN CONTINUED.**MARKS**

- | | | | |
|-----|-------|---|---|
| (c) | (i) | Sketch , on the same set of axes, the graphs of $y = \sin x$ and $y = 1 - \cos x$ over the domain $0 \leq x \leq \pi$. | 2 |
| | (ii) | Write down the values of x for which $\sin x = 1 - \cos x$ in the domain $0 \leq x \leq \pi$. | 1 |
| | (iii) | Evaluate the integral $\int_0^{\pi} (1 - \cos x - \sin x) dx$. | 2 |
| | (iv) | Calculate the area between $y = \sin x$ and $y = 1 - \cos x$ over the domain $0 \leq x \leq \pi$. | 2 |

END OF TEST.

TRIAL MATHEMATICS

SOLUTIONS: 2005

Question 1:

(a) 1.3737... $\leftarrow$ (1)
 1.37 $\leftarrow$ (1) for sig. figs
 (target question for rounding off.)

(b) $x^3 + 2x^2 - 8x$
 $= x(x^2 + 2x - 8) \leftarrow$ (1)
 $= x(x+4)(x-2) \leftarrow$ (1)

(c)(i) $3x^2(4x-1)$
 $= 12x^3 - 3x^2$
 So $\frac{d}{dx}(12x^3 - 3x^2) \leftarrow$ (1)
 $= 36x^2 - 6x \leftarrow$ (1)

OR using the product rule (1)
 $y' = 3x^2 \cdot 4 + (4x-1)6x \leftarrow$ (1)
 $= 12x^2 + 24x^2 - 6x$
 $= 36x^2 - 6x.$

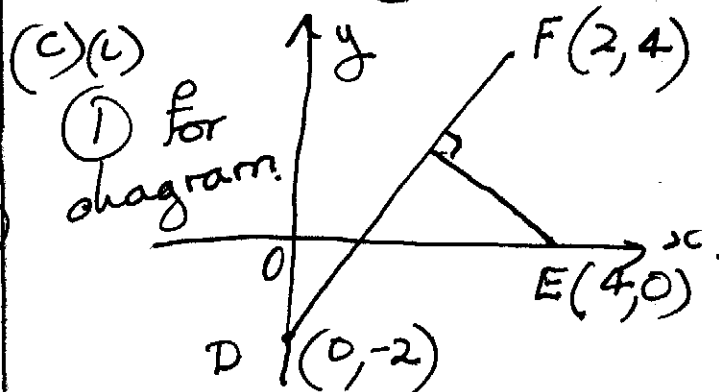
(ii) $\frac{d \log_e x}{dx} = \frac{vu' - uv'}{v^2} \leftarrow$ (1)
 So $\frac{dy}{dx} = \frac{x^3 \cdot \frac{1}{x} - \log_e x \cdot 3x^2}{(x^3)^2}$
 $\leftarrow$ (1)
 $= \frac{x^2 - 3x^2 \log_e x}{x^6}$
 $= \frac{1 - 3 \log_e x}{x^4} \leftarrow$ (1)

(e) $\int \frac{1}{x-1} dx = \ln(x-1) + C$
 (1) $\nearrow$

Question 2:

(a) $L = 5\text{cm}$, $\angle AOB = \frac{\pi}{8}$
 $L = r\theta$
 $S = r \cdot \frac{\pi}{8} \leftarrow$ (1)
 $r = 40/\pi$
 $r = 13\text{cm} \leftarrow$ (1)

(b) let $u = 2^x$
 then $u^2 - 9u + 8 = 0$
 $\leftarrow$ (1) $\rightarrow (u-1)(u-8) = 0$
 $u = 2^x = 1$, $u = 2^x = 8$
 $x = 0 \leftarrow$ (1) $x = 3$



(ii) $DF = \sqrt{(2-0)^2 + (4-2)^2}$
 $= \sqrt{4+36}$
 $= \sqrt{40} \leftarrow$ (1)
 OR $2\sqrt{10}$

(iii) $m = \frac{4 - -2}{2 - 0} = \frac{6}{2} = 3$
 (1) $\nearrow$

$$(iv) y-4=3(x-2)$$

$$y-4=3x-6$$

$$y=3x-2$$

$$3x-y-2=0 \leftarrow (1)$$

$$(v) d = \left| \frac{3 \cdot 4 - 0 - 2}{9+1} \right| \leftarrow (1)$$

$$= \frac{10}{\sqrt{10}} \times \frac{\sqrt{10}}{\sqrt{10}}$$

$$= \sqrt{10} \text{ u}^2 \leftarrow (1)$$

$$(vi) A = \frac{1}{2} \times b \times h$$

$$= \frac{1}{2} \times \sqrt{40} \times \sqrt{10} \leftarrow (1)$$

$$= \frac{1}{2} \sqrt{400}$$

$$= \frac{1}{2} \times 20$$

$$= 10 \text{ u}^2 \leftarrow (1)$$

: Question 3:

$$(a) \frac{d(\tan 3x)}{(3x+2)} \leftarrow (1)$$

$$\frac{dx}{dx} \leftarrow (1)$$

$$= \frac{(3x+2)3\sec^2 3x - \tan 3x \cdot 3}{(3x+2)^2}$$

$$\text{OR}$$

$$= 3 \left[\frac{(3x+2)\sec^2 3x - \tan 3x}{(3x+2)^2} \right]$$

$$(b) \int 3e^{-x/2} dx \leftarrow (1)$$

$$= 3 \int e^{-\frac{1}{2}x} dx$$

$$= 3x - 2e^{-\frac{1}{2}x} + C$$

$$= -6e^{-\frac{1}{2}x} + C \leftarrow (1)$$

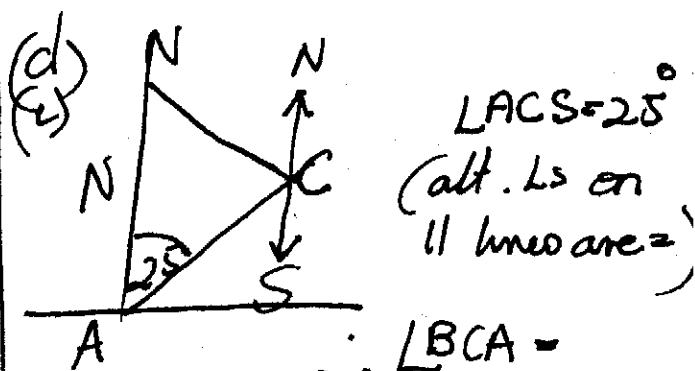
(2)

$$(c) \int_2^4 \frac{3x}{x^2-1} dx$$

$$= \left[\frac{3}{2} \ln(x^2-1) \right]_2^4 \leftarrow (1)$$

$$= \frac{3}{2} [\ln 15 - \ln 3] \leftarrow (1)$$

$$= \frac{3}{2} \ln 5 \leftarrow (1)$$



$$\therefore \angle BCA = 335 - 205 = 130^\circ$$

$$(ii) \frac{d}{\sin 25^\circ} = \frac{5}{\sin 130^\circ} \leftarrow (1)$$

$$d = \frac{5 \times \sin 25}{\sin 130}$$

$$d = 2.8 \leftarrow (1)$$

$$(iii) \triangle ABC \text{ is an isos. } \triangle$$

$$(130 + 25 + \angle ABC = 180^\circ)$$

$$\angle ABC = 25 \leftarrow (1)$$

$$(\angle \text{ sum of } \triangle \text{ from } \angle)$$

$$\text{So total distance is}$$

$$2.8 \times 2 = 5.6 \text{ km}$$

$$= 6 \text{ km} \leftarrow (1)$$

Question 4:

(a) $4\cos x = 1$
 $\cos x = \frac{1}{4}$
 $x = 1.32 \leftarrow \textcircled{1}$
 $x = 2\pi - 1.318 \leftarrow \textcircled{1}$
 $\therefore x = 4.97 \leftarrow \textcircled{1}$

(b) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
 $\textcircled{1} \rightarrow = \left(-\frac{9}{2}\right)^2 - 2x - 2$
 $= 24\frac{1}{4} \leftarrow \textcircled{1}$

(c) 80, 100, 120, ...

(i) $T_{10} = ?$
 an AP where $a = 80$
 $d = 20$ $\leftarrow \textcircled{1}$
 So $T_{10} = 80 + (9)(20)$
 $= 260 \text{ cm.} \leftarrow \textcircled{1}$

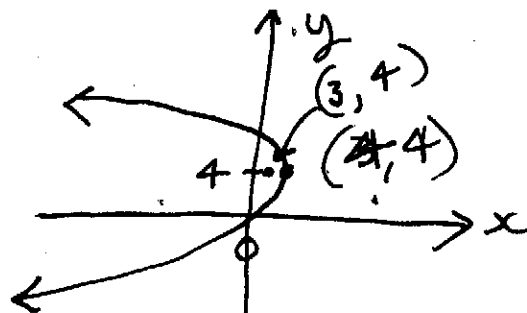
(ii) $\textcircled{1} \rightarrow 450 \geq 80 + 20(n-1)$
 $37 \geq 2n - 2$
 $19\frac{1}{2} \geq n$
 $\therefore \text{to lley} = 19 (\text{integer})$
 $\textcircled{1}$

(d) (i) $4x = 8y - y^2$ $\textcircled{1}$
 $-(4 \times 1) = y^2 - 8y$
 $-4x + 16 = y^2 - 8y + 16$
 $-4(x-4) = (y-4)^2$
 $\therefore \text{Vertex } (4, 4)$
 $\textcircled{1}$

3

(ii) Focus is $(3, 4)$ since $a = 1$ and parabola opens to the left. $\textcircled{1}$

(iii)



$\textcircled{1}$ for sketch & pts.

Question 5:

a(i) $f(x) = 2x^3 - 3x^2 - 36x + 26$

$f'(x) = 6x^2 - 6x - 36$
 Stat. pts when $f'(x) = 0$
 $0 = 6(x^2 - x - 6)$
 $0 = (x-3)(x+2)$
 $x = 3, -2 \leftarrow \textcircled{1}$

$f(3) = 2 \times 3^3 - 3 \times (3)^2 - 36 \times 3 + 26$
 $= -55$

$f(-2) = 2 \times (-2)^3 - 3 \times (-2)^2 - 36 \times (-2) + 26$
 $= 70$

$\therefore$ Stat. pts at $(3, -55)$ &
 $(-2, 70)$
 $\textcircled{1}$ For the y-values.

4

$$y'' = 12x - 6$$

At $x=3$, $y''=30$, >0

$\therefore$ compare up

so relative minimum

$$x = -2$$

$$y'' = -30 < 0$$

$\therefore$ concave down

so rel. max.

(ii) Possible pt. of inflexion

at $y''' = 0$

$$u \quad 12x - 6 = 0$$

$$x = \frac{1}{2}$$

$$\text{So } y = \frac{2x\left(\frac{1}{2}\right)^3 - 3x\left(\frac{1}{2}\right)^2 - 36x\left(\frac{1}{2}\right) + 26}{1}$$

$$y = 7\frac{1}{2}$$

$\therefore$ Pos. pt. of inflexion:

$$\left(\frac{1}{2}, 7\frac{1}{2}\right)$$

Check:

$$\begin{array}{c|c|c|c|} x & 0 & \frac{1}{2} & 1 \\ \hline y'' & -6 & 0 & 6 \end{array} \quad \leftarrow \textcircled{1}$$

-ve +ve

$y'' = 6 / 0 / 6 /$
-ve $\searrow$ / +ve

-ve / +ve

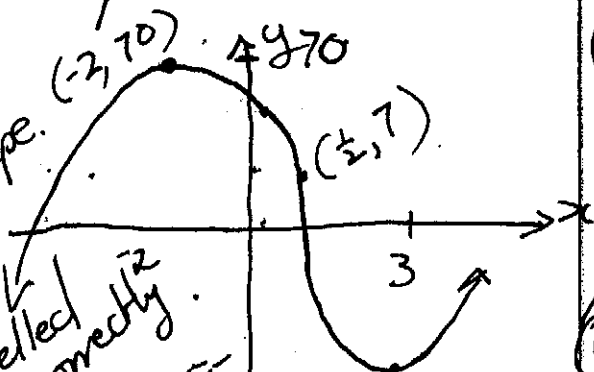
change in concavity

∴ pt. of inflexion.

(iii)

① for shape

① pts labelled correctly. -5



(iv) Concave down when $f''(x) < 0$.

$$u \cdot f''(x) < \frac{1}{2}$$

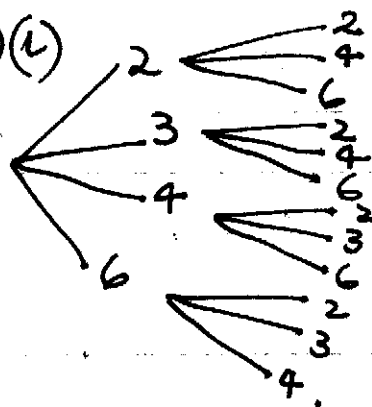
decreasing when

le $-2 < x < 3$

So to satisfy both conditions

$$-2 < x < \frac{1}{2} \leftarrow \textcircled{1}$$

(b)(1)



$$S = 23, 24, 26, 32, 34, 36, 42, 43, 46, 62, 63, 64.$$

② $\rightarrow$ (-1 for each number missing)

$$(ii) P(\text{prime}) = \frac{\frac{2}{12}}{\frac{1}{6}} \leftarrow \textcircled{1}$$

Question 6:

(a) To prove: $\triangle ABC \parallel \triangle AED$

Proof: $\angle DAE = \angle BAC$ $\leftarrow$ (common)

$$\angle AED = \angle ABC = 90^\circ \text{ (Given)}$$

$$\therefore \triangle ABC \parallel \triangle AED$$

(AAA - equiangular)

$$(ii) \frac{AD}{AC} = \frac{AE}{AB}$$

$$\frac{8}{11+AE} = \frac{AE}{10}$$

$$80 = AE^2 + 11AE$$

$$AE^2 + 11AE - 80 = 0$$

$$\text{let } AE = x$$

$$\text{So } (x+16)(x-5) = 0$$

$$x = 5 \text{ or } x = -16$$

$$\text{ie } x = 5 \quad \leftarrow (1)$$

(b) For limiting sum

$$|r| < 1 \quad \leftarrow (1)$$

$$\text{ie } |5 - \sqrt{p}| < 1$$

OR

$$-1 < 5 - \sqrt{p} < 1$$

$$\text{OR } -6 < -\sqrt{p} < -4$$

$$\text{OR } 16 < p < 36 \text{ except}$$

$p \neq 25$
(as if $p=25$ it will not be a series.)

$$(ii) \text{ If } r = 5 - \sqrt{p}$$

$$S_{\infty} = \frac{1}{1 - (5 - \sqrt{20})} \quad \leftarrow (1)$$

$$= \frac{1}{2\sqrt{5} - 4} \times \frac{2\sqrt{5} + 4}{2\sqrt{5} + 4}$$

$$= \frac{2\sqrt{5} + 4}{20 - 16}$$

$$= \frac{\sqrt{5} + 2}{2} \quad \leftarrow (1)$$

(5)

$$(c) y = 2 \tan x$$

$$V = \pi \int_0^{\pi/3} (2 \tan x)^2 dx \quad (1)$$

$$= 4\pi \int_0^{\pi/3} (\sec^2 x - 1) dx \quad \leftarrow (1)$$

$$= 4\pi \left[\tan x - x \right]_0^{\pi/3} \quad \leftarrow (1)$$

$$= 4\pi \left[\tan \frac{\pi}{3} - \frac{\pi}{3} \right] - \left[\tan 0 - 0 \right] \quad \leftarrow (1)$$

$$= 4\pi \left(\sqrt{3} - \frac{\pi}{3} \right) u^3 \quad \leftarrow (1)$$

Question 7:

(a)(i)

x	1	1.5	2	2.5	3
y	1	1.732	3.000	5.196	9.000

(1) $\rightarrow$ (-1 for any number incorrect - don't worry about sig. figs as question is targetted in question (1a).

$$(ii) A = \frac{h}{3} [f(1) + f(3) + 4[f(1.5) + f(2.5)] + 2f(2)] \quad \leftarrow (1)$$

$$A = \frac{1}{3} [(1+9) + 4(1.732 + 5.196) + 2(3)]$$

$$= 7.285 \quad \leftarrow (1)$$

(Q7b)

(i) pts. of intersection

$$-2x = x^2 + 2x - 5$$

$$0 = x^2 + 2x - 5 \leftarrow (1)$$

$$0 = x^2 + 4x - 5$$

$$= (x+5)(x-1)$$

$$x = -5, 1 \leftarrow (1)$$

$$(ii) A = \int_{-5}^1 (-2x - (x^2 + 2x - 5)) dx$$

$$= \int_{-5}^1 (-x^2 - 4x + 5) dx \leftarrow (1)$$

$$\begin{aligned} &= \left[-\frac{1}{3}x^3 - 2x^2 + 5x \right]_{-5}^1 \\ &= \left(-\frac{1}{3} - 2 + 5 \right) - \left(-\frac{125}{3} - 50 - 25 \right) \end{aligned}$$

$$= \frac{8}{3} + \frac{100}{3}$$

$$= 36 \text{ u}^2 \leftarrow (1)$$

(c) $y = e^{2x}$
 $y' = 2e^{2x}$

at $x=1$ $y = e^2$
 $\therefore m = \frac{e^2}{2e^2} = \frac{1}{2}$ $\leftarrow (1)$

So eqn:

$$y - e^2 = \frac{1}{2}(x-1)e^2$$

$$y - e^2 = \frac{1}{2}xe^2 - \frac{1}{2}e^2$$

$$y = \frac{1}{2}xe^2 + \frac{1}{2}e^2 \leftarrow (1)$$

(d) $\sum_{n=2}^4 -2^n$

$$= 2^2 + -2^3 + -2^4 \leftarrow (1)$$

$$= -4 - 8 - 16$$

$$= -28 \leftarrow (1)$$

Question 8

(a) $P(WWW) = \frac{1}{5} \times \frac{1}{5} \times \frac{1}{5}$

$$= \frac{1}{125} \leftarrow (1)$$

(b) $P(WLL) + P(LWL) + P(LLW) + P(WWL) + P(WLW) + P(LWW) + P(WWW)$

$$= 1 - P(LLW) \leftarrow (1)$$

$$= 1 - (.8 \times .8 \times .8)$$

$$= 0.488 \text{ OR } 48.8\% \leftarrow (1)$$

(iii) $P(\text{wins at least 1 game in 'n' games})$
 $= 1 - (.8)^n$ $\leftarrow (1)$

$$\therefore 1 - (.8)^n > .99 \leftarrow (1)$$

$$- (.8)^n > -.01$$

$$(.8)^n < .01 \leftarrow (1)$$

$$\text{So } n(\log .8) < \log(.01)$$

$$\therefore n > \frac{\log(.01)}{\log(.8)}$$

$$n > 20.63$$

$\therefore$ Elvse needs to play 21 games. $\leftarrow (1)$

7

(Q86)

(i) 6 months - \$300, $r = \frac{8}{2} = 4\%$
 $y_1 = 300(1.04) + 300(1.04)^2$
 $= \$636.48 \leftarrow$

(ii) 30 yrs. $60 = n$

Total = $300(1.04 + 1.04^2 + \dots + 1.04^{60})$

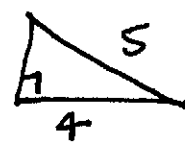
this is a G.P.
 with $a = 1.04$,
 $r = 1.04$
 $n = 60$
 $= 300 \times \frac{1.04(1.04^{60} - 1)}{1.04 - 1}$
 $= \$74253.09 \leftarrow$

(iii) $A_{10} = 300(1.04^{60} + 1.04^{59} + \dots + 1.04^1)$
 next 10 yrs:
 $A_{11} \text{ to } A_{20} = 600(1.04^{40} + \dots + 1.04^1)$
 last 10 yrs
 $A_{21} \text{ to } A_{30} = 1200(1.04^{20} + \dots + 1.04^1)$

$\therefore \text{Total: } 300 + 1.04^1 \left(\frac{1.04 - 1}{.04} \right)$
 $+ 600 \times 1.04^{21} \left(\frac{1.04^{20} - 1}{.04} \right) +$
 $1200 \times 1.04^{41} \left(\frac{1.04^{20} - 1}{.04} \right)$
 $= \$122482.57 \leftarrow$

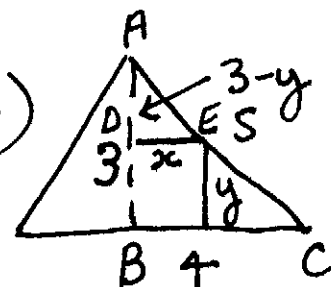
QUESTION NINE:

(a)(i) Using pythagoras:
 and the fact that
 the roof is isosceles



$S^2 - 4^2 = 3^2$

$\therefore$ roof 3m



(ii) $\frac{3}{4} = \frac{3-y}{x}$
 $12 - 4y = 3x$
 $4y = 12 - 3x$
 So $y = \frac{3}{4}(4-x)$

(iii) $V = \pi r^2 h$
 where $x = r$ &
 $h = y = \frac{3}{4}(4-x)$

$V = \pi x^2 y$
 $V = \pi x^2 \cdot \frac{3}{4}(4-x)$
 $= \frac{3\pi}{4}(4x^2 - x^3)$

8

9(a)
(iv)

$$V = \frac{3\pi}{4} (4x^2 - x^3)$$

$$\frac{dV}{dx} = \frac{3\pi}{4} (8x - 3x^2)$$

$$= \frac{3\pi x}{4} (8 - 3x)$$

$\frac{dV}{dx} = 0$ for max.

$$0 = \frac{3\pi x}{4} (8 - 3x)$$

$$x = 0, \frac{8}{3}$$

$x = 0$ cannot exist.
to check if max

$$\frac{d^2V}{dx^2} = \frac{3\pi}{4} (8 - 6x)$$

at $x = \frac{8}{3}$

$$\frac{d^2V}{dx^2} = -18 \cdot \frac{8}{3} < 0$$

$\therefore$ concave down
rel. max.

$\therefore$ Radius for max volume
is $2\frac{2}{3}$ m.

(v) Max vol = $\frac{3\pi}{4} \times \frac{64}{9} \times \frac{4}{3}$

① $\rightarrow \left\{ \begin{aligned} &= \frac{64\pi}{9} \text{ m}^3 \text{ OR} \\ &22.34 \text{ m}^3 \end{aligned} \right.$

(b)

$y = x + m$
 $x^2 + y^2 = 4$
 $x^2 + (x+m)^2 = 4$
 $x^2 + x^2 + 2xm + m^2 = 4$
 $2x^2 + 2xm + m^2 - 4 = 0$

(ii)

$\Delta = 0$
 $2m^2 - 4 \cdot 2(m^2 - 4) = 0$

$$4m^2 - 8m^2 + 32 = 0$$

$$-4m^2 + 32 = 0$$

$$-4m^2 = -32$$

$$m = \pm \sqrt{8}$$

OR $m = 2\sqrt{2}$ and $-2\sqrt{2}$

Question 10:

(i) $\log_e e^{2ax} = 2ax$

OR let $\log_e e^{2ax} = y$

$$\therefore e^y = e^{2ax}$$

$$\therefore y = 2ax$$

(ii) $\int_a^b \log_e e^{2ax} dx = \int_a^b 2ax dx$

$$= \left[ax^2 \right]_a^b$$

$$= ab^2 - aa^2 = ab^2 - a^3$$

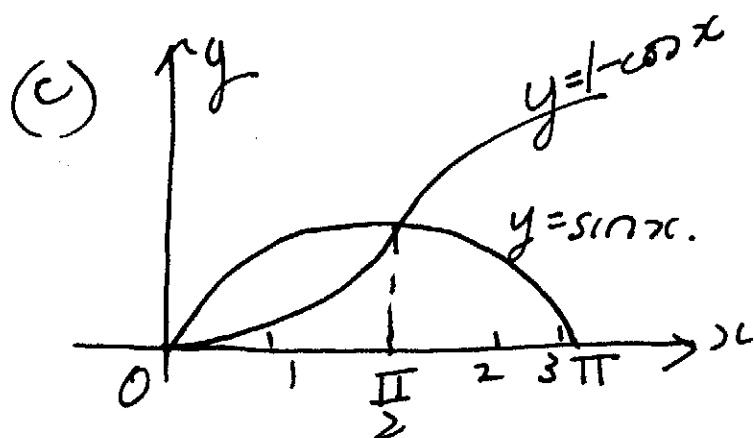
9.

(b)
$$\frac{\cos^3 \theta + \sin \theta \cos \theta}{\sin \theta} =$$

$$\frac{\cos \theta (\cos^2 \theta + \sin^2 \theta)}{\sin \theta} \quad \nwarrow \textcircled{1}$$

$$= \cot \theta$$

So $\cot \theta = 1$ $\textcircled{1}$
 $\therefore \theta = \frac{\pi}{4}, \frac{5\pi}{4}$



$\textcircled{1}$ For each graph.

(ii) $x = \frac{\pi}{2} \neq 0$ $\nwarrow \textcircled{1}$
 (from graph)

(iii)
$$\int_0^{\pi} (1 - \cos x - \sin x) dx$$

$$= \left[x - \sin x + \cos x \right]_0^{\pi} \quad \nwarrow \textcircled{1}$$

$$= \pi - 0 - 1 - (0 - 0 + 1)$$

$$= \pi - 2. \quad \longleftarrow \textcircled{1}$$

(iv)
$$A = \int_0^{\frac{\pi}{2}} (\sin x - (1 - \cos x)) dx$$

$$+ \int_{\frac{\pi}{2}}^{\pi} (1 - \cos x - \sin x) dx$$

$$= \left[-\cos x - x + \sin x \right]_0^{\frac{\pi}{2}} +$$

$$\left[x - \sin x + \cos x \right]_{\frac{\pi}{2}}^{\pi} \quad \nwarrow \textcircled{1}$$

$$= (0 - \frac{\pi}{2} + 1 - (-1)) +$$

$$(\pi - 0 - 1 - (\frac{\pi}{2} - 1)) \quad \nwarrow \textcircled{1}$$

$$= 2 - \frac{\pi}{2} + \frac{\pi}{2}$$

$$= 2 \quad \longleftarrow \textcircled{1}$$

END OF SOLUTIONS.

